

AI-Powered DevOps & SRE Architecture for Finastra LCS TOM

Agentic-based DevOps and SRE for Managed Services that reduce people dependency, improve resilience, reliability, and speed, and align with the **LCS TOM future state**—where AI helps manage infrastructure and application operations within Finastra.

Key Internal Insights from Finastra Resources

From internal documents like [LCS Inhouse Operating Model](#), [Devops SRE 2.0](#), and LCS Roadmap v1.0, here are the major themes: [\[Devops SRE 2.0 | PowerPoint\]](#), [\[LCS Inhouse...ting Model | PowerPoint\]](#), [\[LCS Roadmap v1.0 | PowerPoint\]](#)

1. Why Agentic DevOps & SRE?

- Current SDLC has limited cloud adoption; integrating **cloud-native principles early** improves agility and scalability.
- Moving from manual operations to **touchless automation** and **observability-driven remediation** is critical for reducing operational toil.

2. Future State Vision for LCS TOM

- **AI-driven service management:** Self-healing (AIOps), proactive monitoring, and predictive maintenance.
- **Zero downtime upgrades, standardized experience, and self-service capabilities.**
- KPI targets: **99.95% uptime, 95–100% cloud-native adoption, and automation-first deployment.**

3. Strategic Levers

- **DevOps as a Service (DaaS)** for internal and partner ecosystems.
- **CI/CD pipelines** for infrastructure, application deployment, and database management using tools like GitHub, ADO, Terraform, and Ansible ([SRE LCS](#)).

- **Cross-functional teams** and simplified RACI models to reduce coordination overhead.

4. Cost & Efficiency Goals

- Reduce TCO from ~\$500K to ~\$250K annually for KCS offers.
 - Lower sales cycle by 50% and double win rates through automation and standardization.
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Industry Best Practices & Trends

External sources highlight how **agentic AI** and **AIOps** are transforming DevOps and SRE: [\[techcommu...rosoft.com\]](#), [\[infoworld.com\]](#), [\[linkedin.com\]](#), [\[fulcrumdigital.com\]](#), [\[aws.amazon.com\]](#)

- **Agentic AI in Operations:**
 - AI agents act autonomously to interpret context, trigger workflows, and remediate issues without human intervention.
 - Example: Azure SRE Agent automates incident response, certificate rotation, and scaling across hybrid environments.
 - **AIOps for Predictive & Self-Healing Infrastructure:**
 - Event correlation, anomaly detection, and proactive remediation reduce MTTR by up to 60%.
 - AI-governed infrastructure enables **self-healing**, **auto-scaling**, and **continuous compliance**.
 - **Benefits for Managed Services:**
 - **Zero-touch operations:** AI agents handle repetitive tasks, freeing engineers for strategic work.
 - **Improved reliability & resilience:** Predictive analytics prevent outages before they occur.
 - **Faster time-to-market:** Automated CI/CD pipelines and intelligent orchestration accelerate deployments.
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How to Make It Robust for LCS TOM

Here's a **structured roadmap** for Finastra:

1. Build an Agentic DevOps Framework

- Integrate **AI agents** into CI/CD pipelines for infrastructure and application deployments.
- Use **Model Context Protocol (MCP)** and OpenAPI for seamless orchestration across Azure and internal systems.

2. Embed AIOps for Autonomous Operations

- Implement **observability platforms** with anomaly detection and predictive alerts.
- Enable **self-healing workflows** for common failure scenarios (e.g., container crashes, certificate expiry).

3. Standardize & Automate Across Value Streams

- Adopt **Infrastructure as Code (IaC)** for all environments.
- Automate post-deployment checks and vulnerability remediation.

4. Cultural & Organisational Shift

- Promote a **DevOps/SRE mindset**: shared responsibility, continuous learning, and proactive incident management.
- Establish **cross-functional squads** with AI-assisted decision-making.

5. Governance & Compliance

- Ensure AI-driven operations comply with ISO 27001, CMMI Level 4/5, and financial industry standards.
 - Implement **AI governance** for transparency and auditability.
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1. Executive Summary

An AI-driven DevOps & SRE operating model enabling self-healing, resilient, faster, and lower-TCO operations for LCS (Loan IQ, TI, CCB, Nexus). Built around agentic AI, cloud-native automation, observability, and standardized CI/CD.

2. Reference Architecture

Experience & Control Plane

- Teams/Adaptive Cards
- Jira/ServiceNow integration
- SRE/Exec Cockpits (SLOs, reliability, risk)

Agentic Ops Layer (AIOps)

- Autonomous agents for incidents, remediation, releases, security
- Runbook graph (no-code + coded automation)
- Policy engine (SLO/error-budget governance)
- MCP/OpenAPI connectors to cloud, CI/CD, observability, ITSM

DevOps Platform

- GitHub/ADO repositories & pipelines
- IaC (Terraform/Bicep), Config-as-Code (Helm/Kustomize)
- [SAST/DAST](#)/Container/IaC scanning

Azure Runtime Platform

- Azure Front Door (WAF)
- Private Link → AKS (Istio ingress)
- PaaS databases, messaging, storage
- Key Vault + Managed Identity
- Multi-tenant isolation (RG/Subscription/Namespace)

Observability & Security

- Prometheus, Grafana, OpenTelemetry
 - Azure Monitor, Log Analytics
 - Defender for Cloud, Aqua, Sentinel
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3. Design Decisions

- Prefer AFD → Private Link → AKS for minimal latency
 - First-class agentic runbooks for top operational tasks
 - SLO/error-budget-driven release gating
 - DevSecOps embedded into pipelines
 - Progressive delivery (canary, blue/green) with automated rollback
 - Continuous compliance and evidence capture
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4. Agentic Operating Model

Incident Auto-Triage & Self-Healing

- AI correlates metrics/logs/traces
- Identifies blast radius
- Executes remediation (restart pods, scale, certificate rotation)

Vulnerability Remediation

- Auto-triage Defender/Aqua findings
- Auto-generate PRs for base image bumps
- Run gated rollout & attach evidence to ITSM

Release Guardrails

- Automated BoM generation, risk scoring, chaos tests
- Progressive rollout driven by agent checks

Certificate & Secret Rotation

- Auto-renewal via Key Vault
 - Propagated to AFD/Istio with validation
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5. CI/CD Strategy

- GitOps-aligned pipelines

- Build → Scan → Sign → Deploy → Post-deploy checks
 - Promotion path: DH Tenant → Internal LCS → IBM Tenant → Customers
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6. Observability & SRE Practices

- Standardized SLIs (availability, latency, error rate, saturation)
 - Error budgets controlling release velocity
 - Unified tracing across all services
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7. Security, Compliance & Governance

- Landing zone policy enforcement
 - Auto-evidence collection for ISO/CMMI audits
 - RBAC/ABAC for telemetry and operational actions
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8. Multi-Tenancy & DR

- Per-tenant RG/subscription isolation
 - Region DR (active/standby), automated DR runbooks
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9. Implementation Roadmap

Phase 1 – Foundation

- Golden IaC baselines, observability standards
- 5–10 high-value agentic runbooks
- Define product SLOs/SLIs

Phase 2 – Scale Out

- Full runbook catalog (vuln remediation, capacity)
- Progressive delivery across rings
- Integrate SecOps signals into agentic workflows

Phase 3 – Autonomous Ops

- Self-healing for known failure modes
 - Automated DR exercises
 - Continuous compliance and FinOps optimization
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10. KPIs & Expected Outcomes

- $\geq 99.95\%$ availability
 - Reduced MTTR via AI triage
 - Faster deployments, fewer change failures
 - 30–40% operational cost reduction potential
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11. Example Agentic Runbook (YAML)

YAML ▾

```
1 name: liq-api-latency-mitigation
2 trigger:
3   type: anomaly
4   condition:
5     metric: http_latency_p95_ms
6     operator: ">"
7     threshold: 800
8 actions:
9   - detect_blast_radius
10  - scale_out: { replicas: +2 }
11  - istio_route_shift: { blue: 70 }
12  - verify:
13      checks:
14        - latency < 600 for 10m
15        - error_rate < 0.5%
16  - resume_rollouts
17  - open_post_incident
18
```

